Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

Task – 2

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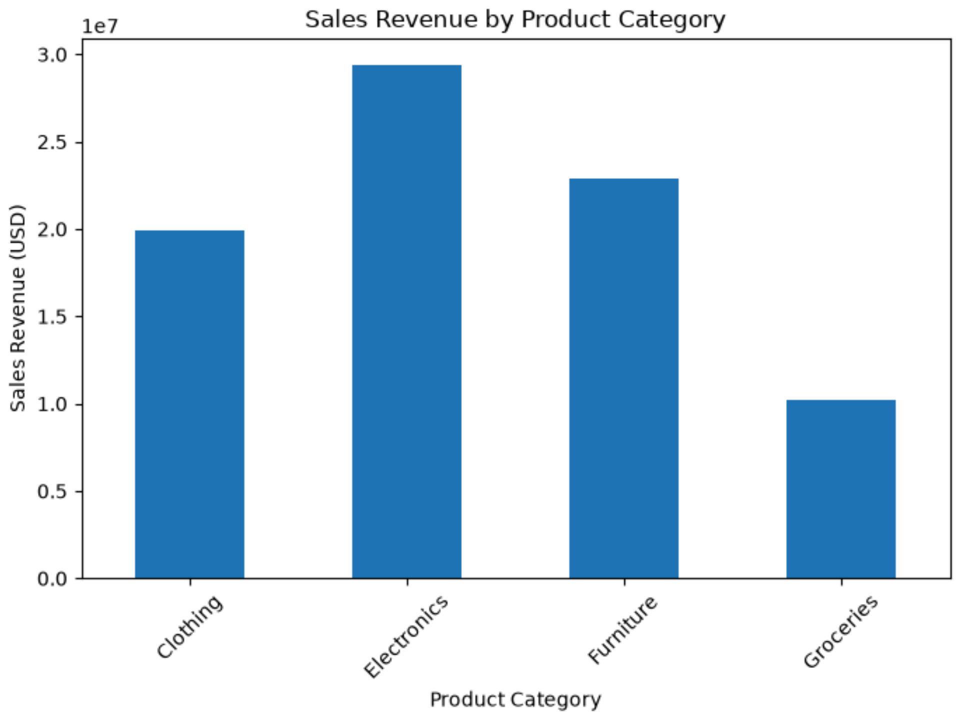
Extra Task:
Perform following questions using Retail.csv
Q.1 Write a Python program to:
Count the number of products in each Product Category.
Find the category with the highest sales.
Display the result using a bar chart.
Code & Output:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("Retail_sales.csv")

# Count products in each Product Category
product_count = df["Product Category"].value_counts()
print("Number of Products in Each Category:")
print(product_count)

# Total sales by Product Category
category_sales = df.groupby("Product Category")["Sales Revenue (USD)"].sum()

# Category with highest sales
highest_category = category_sales.idxmax()
print("\nCategory with Highest Sales:", highest_category)
```

```
Number of Products in Each Category:
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

```
# Bar Chart
plt.figure(figsize=(8,5))
category_sales.plot(kind="bar")
plt.title("Sales Revenue by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Sales Revenue (USD)")
plt.xticks(rotation=45)
plt.show()
```



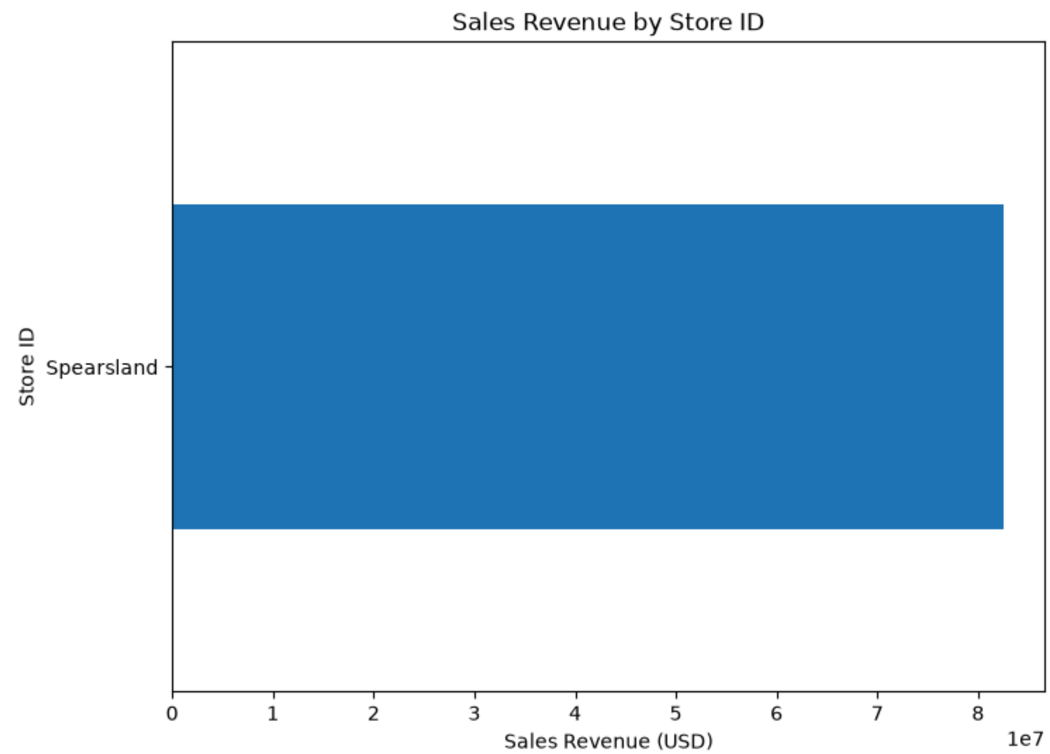
Q.2 Write a Python program to:
Calculate the total Sales Revenue for each Store ID.
Find the best-performing store.
Find the least-performing store.
Display the result using a horizontal bar chart.

Code & Output:

```
# Total Sales Revenue by Store ID
store_sales = df.groupby("Store ID")["Sales Revenue (USD)"].sum()

print("Total Sales Revenue for Each Store:")
print(store_sales)

# Best-performing Store
print("\nBest-performing Store:", store_sales.idxmax())

# Least-performing Store
print("Least-performing Store:", store_sales.idxmin())
```

```
# Horizontal Bar Chart
plt.figure(figsize=(8,6))
store_sales.plot(kind="barh")
plt.title("Sales Revenue by Store ID")
plt.xlabel("Sales Revenue (USD)")
plt.ylabel("Store ID")
plt.show()
```

```
Total Sales Revenue for Each Store:
Store ID
Spearsland    82485287.78
Name: Sales Revenue (USD), dtype: float64

Best-performing Store: Spearsland
Least-performing Store: Spearsland
```

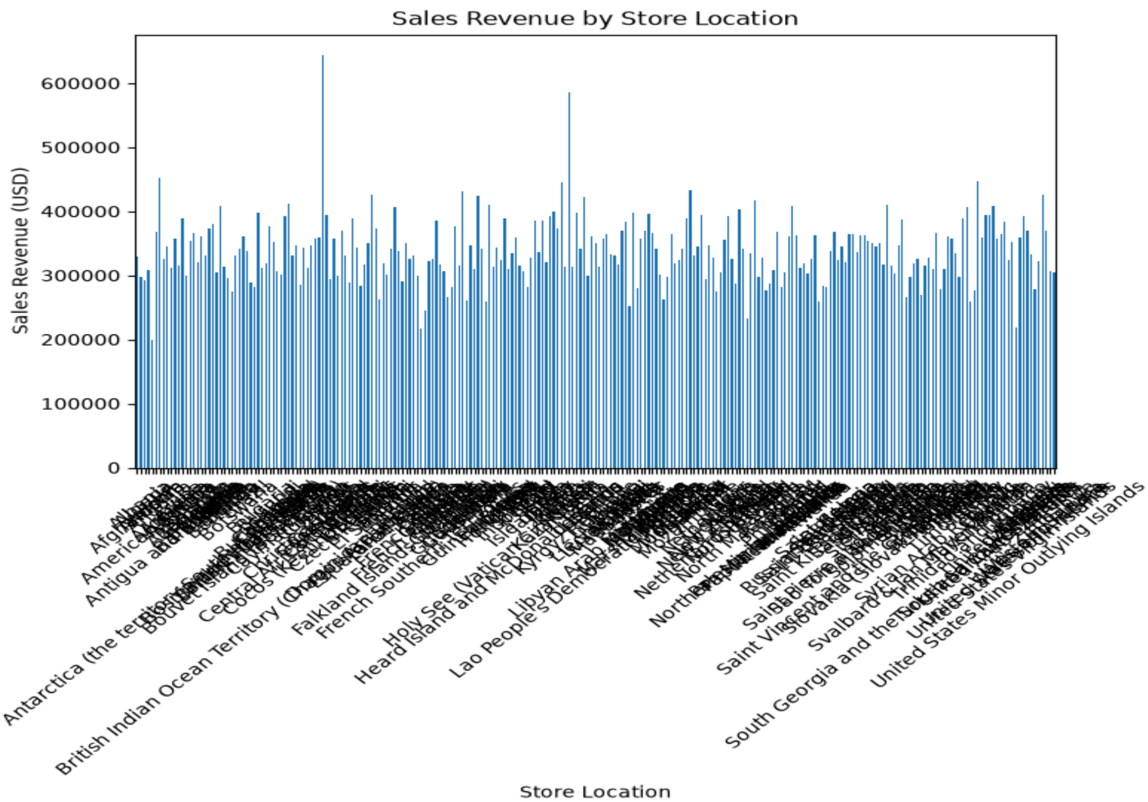


Q.3 Write a Python program to:
Calculate total Sales Revenue for each Store Location.
Identify the location with the highest sales.
Plot the results using a bar chart.
Code & Output:

```
# Total Sales Revenue by Store Location
location_sales = df.groupby("Store Location")["Sales Revenue (USD)"].sum()

print("Sales Revenue by Store Location:")
print(location_sales)

# Highest Sales Location
print("\nLocation with Highest Sales:", location_sales.idxmax())
```

```
# Bar Chart
plt.figure(figsize=(8,5))
location_sales.plot(kind="bar")
plt.title("Sales Revenue by Store Location")
plt.xlabel("Store Location")
plt.ylabel("Sales Revenue (USD)")
plt.xticks(rotation=45)
plt.show()
```

```
Sales Revenue by Store Location:
Store Location
Afghanistan         329511.22
Albania             296929.47
Algeria             292690.46
American Samoa      308614.68
Andorra             199063.87
                      ...
Wallis and Futuna   323061.14
Western Sahara      426248.78
Yemen               369518.90
Zambia              307427.30
Zimbabwe            305188.53
Name: Sales Revenue (USD), Length: 243, dtype: float64
```



Q.4 Calculate the following descriptive statistics for the Units Sold column:
Mean, Median, Mode, Minimum, Maximum, Range, Variance, Standard Deviation

Code & Output:

```python
units = df["Units Sold"]

print("Mean:", units.mean())
print("Median:", units.median())
print("Mode:", units.mode()[0])
print("Minimum:", units.min())
print("Maximum:", units.max())
print("Range:", units.max() - units.min())
print("Variance:", units.var())
print("Standard Deviation:", units.std())
```

✓ 0.0s

```
Mean: 6.161966666666666
Median: 6.0
Mode: 7
Minimum: 0
Maximum: 56
Range: 56
Variance: 11.048501748947187
Standard Deviation: 3.323928661831837
```

Perform following operations on Netflix.csv

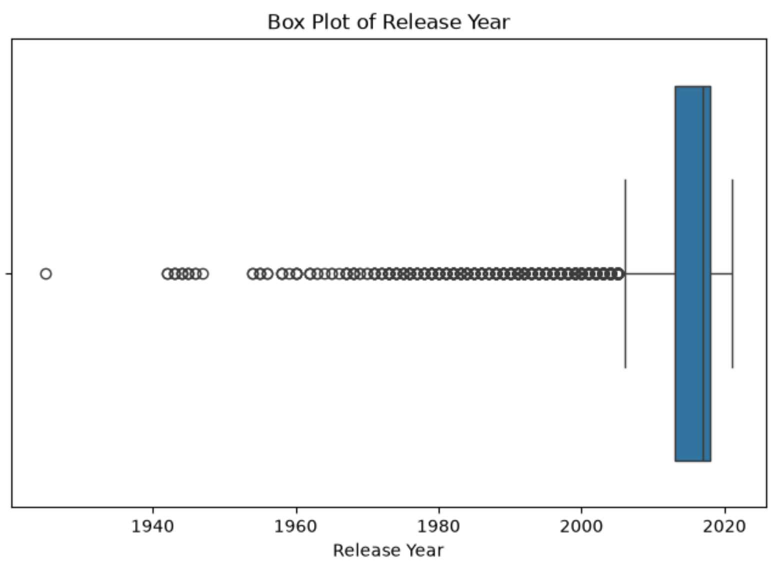
Q.1 Create a Box Plot for the Release Year column.

Identify: Minimum value, Maximum value, Median Presence of outliers

Code & Output:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("NetFlix - NetFlix.csv")
# Box Plot
plt.figure(figsize=(8, 5))
sns.boxplot(x=df["release_year"])
plt.title("Box Plot of Release Year")
plt.xlabel("Release Year")
plt.show()
```

✓ 0.1s



Q.2 Filter only Movies from the dataset.

Extract the numerical value from the Duration column (e.g., 90 min → 90).

Calculate:Mean, Median, Mode, Variance, Standard Deviation for movie duration.

Code & Output:

```
# Minimum, Maximum and Median
print("Minimum Value :", df["release_year"].min())
print("Maximum Value :", df["release_year"].max())
print("Median :", df["release_year"].median())
```

✓ 0.0s

```
Minimum Value : 1925
Maximum Value : 2021
Median : 2017.0
```

```
# Outliers using IQR
Q1 = df["release_year"].quantile(0.25)
Q3 = df["release_year"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[(df["release_year"] < lower) |
              (df["release_year"] > upper)]
print("Number of Outliers :", len(outliers))
print("Outliers are present :", len(outliers) > 0)
```

✓ 0.0s

```
Number of Outliers : 745
Outliers are present : True
```